Design Project CEE 181

Team:

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Question

On a three-lane freeway with an on-ramp and a left-lane work-zone closure, when does traffic at the merge stay free-flowing, and when does it form a persistent queue? And How do model assumptions change the answer?

Prediction

At the base demand of 1,800 mainline + 300 ramp veh/h, the closure causes only local slowing. During the peak of 3,000 + 450 veh/h (240-480 s), merging disturbances at the closure reduce discharge. A queue then forms upstream of 650 m and grows during the peak. Under identical demand without the closure, no queue forms.

Model and sources

Vehicles follow the Intelligent Driver Model (IDM) [1],

$$dv/dt = a \times [1 - (v/v0)^4 - (s^*/s)^2], \quad s^* = s0 + \max(0, vT + v\Delta v / (2\sqrt{ab}))$$

Here s is the bumper-to-bumper gap and $\Delta v = v - v_{leader}$. All accelerations are computed from the same snapshot before positions update.

Parameter	Value
v0	25–31 m/s (5 levels by ID)
T / s0	1.5 s / 2 m
a / b	1.2 / 2.0 m/s²
Time step	0.05 s, ballistic update

Lane changes use a simplified MOBIL rule [3] with gap (≥ 1.5 m) and braking (≥ −3 m/s^2) safety checks. Followers yielding to merging cars is our own assumption.

Segment: 1,000m. The ramp joins at 250m and the left lane is closed from 650 to 850 m. Detectors at 200, 550 and 900 m report every 30s.

Limitations:

- Parameters are uncalibrated.
- The closure is a stopping obstacle, not a tapered lane.
- Boundary storage is unlimited, so upstream spillback is not modeled.

The minimum example is three paired 600 s runs with identical arrivals.

Run	Exits	Delay (veh·s)	Max. queue (veh)	Ramp queue at 600s
Open road	417	1,268	0	0
Closure	385	7,902	33	0
Closure + fixed metering	371	7,772	38	12

The closure raises delays about sixfold and creates a queue of up to 33 vehicles; the open road has none. Fixed metering (one release per 12s) cuts delay by under 2% and leaves 12 cars on the ramp.

Checks passing

- vehicle conservation every step
- no overlaps, negative speeds or closed-lane entries
- IDM stopping and equilibrium tests
- per-lane detector counts
- halving the time step changes delay by about −2.9%

Interfaces

Units are SI. Position is measured from the upstream boundary.

Data	File	Fields
Vehicle state	traces.csv	time_s, vehicle_id, lane_id, position_m, speed_mps
Detector	observations.csv	sensor_id, interval_start_s, interval_end_s, count, mean_speed_mps, quality_flag
Control	control_log.csv	issue_time_s, actuator_id, action, reason

Final pathways considered

- **A: CAV.** ACC/CACC vehicles to smooth merging. This reuses the IDM code.
- **B: Signal control.** A queue-responsive signal compared with a fixed-time plan. A prototype exists

Next steps

- Run until the queue clears.
- Test sensitivity to the yielding rule.
- Run a lead-vehicle braking test.
- Compare with a second model.

References

1. Treiber, Hennecke & Helbing (2000). *Physical Review E* 62, 1805–1824. https://doi.org/10.1103/PhysRevE.62.1805
2. Treiber, M. IDM documentation. https://mtreiber.de/MicroApplet/IDM.html
3. Kesting, Treiber & Helbing (2007). *Transportation Research Record* 1999, 86–94. https://doi.org/10.3141/1999-10
4. FHWA *Traffic Signal Timing Manual*, Ch. 5. https://ops.fhwa.dot.gov/publications/fhwahop08024/chapter5.htm